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Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\jakka>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 28
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database library_db;
Query OK, 1 row affected (0.14 sec)

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```

mysql> show databases;
+-----+
| Database |
+-----+
| bookflow_db |
| information_schema |
| library_db |
| mysql |
| performance_schema |
| sys |
| usersdb |
+-----+
7 rows in set (0.09 sec)

mysql> use library_db;
Database changed

```

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mysql> create table books(book_id int,title varchar(100),isbn varchar(20),published_year int check (published_year < 2027));
Query OK, 0 rows affected (0.12 sec)

mysql> insert into books values(1, 'The Great Gatsby', '9780743273565', 1925);
Query OK, 1 row affected (0.06 sec)

mysql> insert into books values(2, 'To Kill a Mockingbird', '9780061120084', 1960);
Query OK, 1 row affected (0.05 sec)

mysql> insert into books values(3, '1984', '9780451524935', 1949);
Query OK, 1 row affected (0.06 sec)

mysql> select * from books;
+-----+-----+-----+-----+
| book_id | title | isbn | published_year |
+-----+-----+-----+-----+
| 1 | The Great Gatsby | 9780743273565 | 1925 |
| 2 | To Kill a Mockingbird | 9780061120084 | 1960 |
| 3 | 1984 | 9780451524935 | 1949 |
+-----+-----+-----+-----+
3 rows in set (0.04 sec)

```

```
mysql> create table members(member_id int,full_name varchar(100),email varchar(100));
Query OK, 0 rows affected (0.08 sec)

mysql> insert into members values(101, 'John Smith', 'john.smith@email.com');
Query OK, 1 row affected (0.06 sec)

mysql> insert into members values(102, 'Emma Wilson', 'emma.wilson@email.com');
Query OK, 1 row affected (0.06 sec)

mysql> insert into members values(103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 1 row affected (0.05 sec)

mysql> select * from members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> create table loans(loan_id int,member_id int,book_id int,loan_date date);
Query OK, 0 rows affected (0.12 sec)

mysql> insert into loans values(1, 101, 1, '2025-01-05');
Query OK, 1 row affected (0.09 sec)

mysql> insert into loans values (2, 102, 2, '2025-01-08');
Query OK, 1 row affected (0.03 sec)

mysql> insert into loans values(3, 103, 3, '2025-01-10');
Query OK, 1 row affected (0.05 sec)

mysql> insert into loans values(4, 101, 2, '2025-02-01');
Query OK, 1 row affected (0.05 sec)

mysql> insert into loans values(5, 102, 1, '2025-02-05');
Query OK, 1 row affected (0.01 sec)

mysql> insert into loans values (6, 103, 2, '2025-02-12');
Query OK, 1 row affected (0.06 sec)

mysql> insert into loans values (7, 101, 3, '2025-03-01');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> select * from loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

10 rows in set (0.00 sec)

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title  
-> FROM loans l  
-> INNER JOIN members m ON l.member_id = m.member_id  
-> INNER JOIN books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Great Gatsby
Michael Brown	The Great Gatsby
Emma Wilson	The Great Gatsby
John Smith	The Great Gatsby
Michael Brown	To Kill a Mockingbird
John Smith	To Kill a Mockingbird
Emma Wilson	To Kill a Mockingbird
Emma Wilson	1984
John Smith	1984
Michael Brown	1984

10 rows in set (0.00 sec)

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books  
-> FROM books  
-> GROUP BY published_year  
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1

3 rows in set (0.05 sec)

```
mysql> CREATE TABLE donation_history (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES books(book_id)
-> );
Query OK, 0 rows affected (0.18 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO donation_history (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.07 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.05 sec)

mysql> select * from donation_history;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1 | 4 | Raj Kumar | 2026-07-30 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> select * from books;
+-----+-----+-----+-----+
| book_id | title | isbn | published_year |
+-----+-----+-----+-----+
| 1 | The Great Gatsby | 9780743273565 | 1925 |
| 2 | To Kill a Mockingbird | 9780061120084 | 1960 |
| 3 | 1984 | 9780451524935 | 1949 |
| 4 | Animal Farm | 9780451526342 | 1945 |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> CREATE INDEX idx_books_isbn ON books(isbn);
Query OK, 0 rows affected (0.18 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM books WHERE isbn = '9780451524935';
+-----+-----+-----+-----+
| book_id | title | isbn | published_year |
+-----+-----+-----+-----+
| 3 | 1984 | 9780451524935 | 1949 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```